

Replicating Rust (1987): Dynamic Bus Engine Replacement

Team Members: Zheshi Li and Tongfang Tian

Project Summary

Significance. As we explore to some discrete choice model (regression) like logit and Poisson model in class. They are all static. We push ourselves to consider a dynamic model. Our project will replicate Rust's classic 1987 paper on bus engine replacement, one of the foundational papers in structural econometrics and dynamic discrete choice modeling. The paper is especially interesting because it turns a very practical maintenance decision into a rigorous forward looking, infinite horizontal optimization problem: a bus manager must decide whether to replace an engine now or continue operating it and risk higher future maintenance costs and breakdowns. This topic is important not only because of its clear economic intuition, but also because Rust's framework became a benchmark for later research in industrial organization, labor economics, and empirical microeconomics more broadly. For us, the project is valuable because it provides a concrete way to understand how dynamic programming, state transitions, and likelihood-based estimation work together in a structural model. For others, it remains a useful example of how economists recover decision rules and cost parameters from observed choices rather than estimating only reduced-form model (Regression).

Outcomes. The main outcome of the project is to reproduce the core empirical and computational results of Rust's bus engine replacement model. In particular, we aim to recover the key structural estimates for replacement cost and maintenance cost, replicate the comparison between the forward-looking specification and the myopic specification, and reproduce the main likelihood-based evidence that the bus manager was not behaving myopically. We also plan to reproduce at least one of the main figures, such as the replacement probability or hazard curve over mileage states, in order to show the economic implications of the estimated model rather than only reporting coefficients.

Techniques and Software to Be Used. There are already some existing replication tools for Rust's model, such as Ferrall's `niqlow` files and the Python package `ruspy`. These packages are useful references because they show how the Rust model can be organized and estimated with modern computational tools. However, instead of relying directly on those packages, we plan to write our own code in Python so that we can understand the main computational steps of the model more clearly. Our project will focus on two standard estimation methods in the dynamic discrete choice literature. First, we will implement Rust's Nested Fixed Point (NFP) method, which solves the dynamic programming problem inside the likelihood estimation. This is the method used in the original paper, and it is important because it shows directly how the Bellman equation, the value function, and the structural parameters fit together. Second, we will implement the conditional choice probability (CCP) method developed by Hotz and Miller (1993). Because NFP is complex and computational intensive. Hotz and Miller's method is more easy in computation, using observed choice probabilities to recover the relative value of decisions, which can avoid solving the full dynamic programming problem at every parameter guess (every inner loop). We expected the second method would be quicker. In this sense, it provides a useful comparison with the NFP approach and helps illustrate an important later development in the dynamic discrete choice

literature. In practice, we will use Python for data cleaning, state construction, estimation, and plotting. The main steps will include organizing Rust's bus data, constructing the mileage states, estimating transition probabilities, coding value function iteration for the NFP approach, and then implementing the CCP estimator as an alternative method. By working through both approaches, we hope to better understand the computational trade-offs between the original Rust method and later alternatives. Our final version will report the main estimates and figures and discuss what we learned from reproducing the model with these two different estimation strategies.